Homework 5

MATH 441: DIFFERENTIAL EQUATIONS

NAME: DATE:

(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

1. Exercise §3.5 #9. Find the general solution of the given differential equation

$$2y'' + 3y' + y = t^2 + 3\sin t$$

2. Exercise §3.5 #17. Find the solution of the given intial value problem.

$$y'' - 2y' + y = te^t + 4, \qquad y(0) = 1, \qquad y'(0) = 1$$

In each of Problems 3 and 4, use the method of variation of parameters to find a particular solution of the given differential equation. Then check your answer using the method of undetermined coefficients.

3. Exercise §3.6 #1.

$$y'' - 5y' + 6y = 2e^t$$

4. Exercise §3.6 #3.

$$y'' + 2y' + y = 3e^{-t}$$

In each of Problems 5 through 7, find the general solution of the given differential equations.

5. Exercise §4.2 #13.

$$2y''' - 4y'' - 2y' + 4y = 0$$

6. Exercise §4.2 #14.

$$y^{(4)} - 4y''' + 4y'' = 0$$

7. Exercise §4.3 #4.

$$y''' - y' = 2\sin t$$